

Some Details

Su Yingze

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1 Normalization of Polarization Vectors

First, we give the definition of polarization vectors which is written in terms of adjoint representation

$$\varepsilon_a^\mu = \frac{\langle p_I | (r \cdot \Gamma) \Gamma^\mu | p^J \rangle (\sigma_a)^I_J}{4(p \cdot r)} \quad (1)$$

here a runs from 1 to 3. We can use antisymmetric tensor to rewrite this equation, like

$$\varepsilon_a^\mu = \frac{\langle p_I | (r \cdot \Gamma) \Gamma^\mu | p_K \rangle \epsilon^{JK} (\sigma_a)^I_J}{4(p \cdot r)} \quad (2)$$

here we choose the left-multiplication convention.

Then, a 2×2 symmetric matrix can be defined like

$$(\sigma_a)^{IJ} = \epsilon^{JK} (\sigma_a)^I_K. \quad (3)$$

This means that the original polarization vector can be understood as

$$\begin{aligned} \varepsilon_a^\mu &= -\frac{\langle p_I | (r \cdot \Gamma) \Gamma^\mu | p_J \rangle}{4(p \cdot r)} (\sigma_a)^{IJ} \\ &= -\varepsilon_{IJ}^\mu (\sigma_a)^{IJ} \end{aligned} \quad (4)$$

here I, J are fundamental little group indices, runs from 1 to 2.

And we know that the reasonable polarization vector for physical state should be symmetric with little group indices, it means that we need to symmetrize the indices I and J as $(IJ) = \frac{1}{2}(IJ + JI)$. Because of the symmetry of $(\sigma_a)^{IJ}$, so it can be obtained that

$$\varepsilon_a^\mu = -(\varepsilon_{(IJ)}^\mu + \varepsilon_{[IJ]}^\mu) (\sigma_a)^{IJ} = -\varepsilon_{(IJ)}^\mu (\sigma_a)^{IJ} \quad (5)$$

which means we only need to consider the symmetric part of ε_{IJ}^μ .

Now, we want to compute the normalization for ε_{IJ}^μ , and after multiplying the symmetric matrix, it should go back to the normalization

$$\varepsilon_a^\mu \varepsilon_{b,\mu} = -\delta_{ab} \quad (6)$$

The method to compute the normalization is just multiplying two polarization vectors directly, like

$$\varepsilon_{(IJ)}^\mu \varepsilon_{(KL),\mu} = \frac{\langle p_I | (r \cdot \Gamma) \Gamma^\mu | p_J \rangle}{4(p \cdot r)} \times \frac{\langle p_K | (r \cdot \Gamma) \Gamma_\mu | p_L \rangle}{4(p \cdot r)} \quad (7)$$

here we need to use a special 5d equality

$$(\Gamma^\mu)_A^B (\Gamma_\mu)_C^D = -2\Omega_{AC} \Omega^{BD} + 2\delta_A^D \delta_C^B - \delta_A^B \delta_C^D \quad (8)$$

and the dirac operator can be decomposed to

$$\not{p} = p \cdot \Gamma = p_A^B = |p_I\rangle_A \langle p^I|^B \quad (9)$$

which help us to rewrite the equation (7) to be

$$\begin{aligned} \varepsilon_{IJ}^\mu \varepsilon_{KL,\mu} &= \frac{\langle p_I |^A |r_M\rangle_A \langle r^M |^B (\Gamma^\mu)_B^C |p_J\rangle_C \langle p_K |^D |r_N\rangle_D \langle r^N |^E (\Gamma_\mu)_D^F |p_L\rangle_F}{16(p \cdot r)^2} \\ &= \frac{\langle p_I |r_M\rangle \langle p_K |r_N\rangle}{16(p \cdot r)^2} \times \langle r^M |^B (\Gamma^\mu)_B^C |p_J\rangle_C \langle r^N |^E (\Gamma_\mu)_E^F |p_L\rangle_F \end{aligned} \quad (10)$$

After plugging the equation (8) into the second part of (10), it becomes

$$\begin{aligned} &\langle r^M |^B |p_J\rangle_C \langle r^N |^E |p_L\rangle_F (-2\Omega_{BE} \Omega^{CF} + 2\delta_B^F \delta_E^C - \delta_B^C \delta_E^F) \\ &= -2\langle r^M |r^N\rangle \langle p_L |p_J\rangle + 2\langle r^M |p_L\rangle \langle r^N |p_J\rangle - \langle r^M |p_J\rangle \langle r^N |p_L\rangle \end{aligned} \quad (11)$$

The first term of (11) equals to 0 because the equality

$$\langle p_I |p_J\rangle = 0 \quad (12)$$

and the remaining two term can be contracted with the first part of (10), which helps to obtain the following two terms

$$2\langle p_I |r_M\rangle \langle r^M |p_L\rangle \langle p_K |r_N\rangle \langle r^N |p_J\rangle - \langle p_I |r_M\rangle \langle r^M |p_J\rangle \langle p_K |r_N\rangle \langle r^N |p_L\rangle \quad (13)$$

It is known that

$$\langle p_I | (r \cdot \Gamma) | p_J \rangle = \text{Tr}[|p_J\rangle \langle p_I | (r \cdot \Gamma)] = \frac{1}{2} \epsilon_{IJ} \text{Tr}[(p \cdot \Gamma)(r \cdot \Gamma)] = 2\epsilon_{IJ}(p \cdot r) \quad (14)$$

so the formal equation can be rewritten like

$$8(p \cdot r)^2 \epsilon_{IL} \epsilon_{KJ} - 4(p \cdot r)^2 \epsilon_{IJ} \epsilon_{KL} \quad (15)$$

Finally, we need to symmetrize the indices (IJ) and (KL) , so that we obtain

$$\varepsilon_{IJ}^\mu \varepsilon_{KL,\mu} = \frac{1}{8(p \cdot r)^2} (p \cdot r)^2 (\epsilon_{IL} \epsilon_{KJ} + \epsilon_{JL} \epsilon_{KI} + \epsilon_{IK} \epsilon_{LJ} + \epsilon_{JK} \epsilon_{LI}) - \frac{1}{16(p \cdot r)^2} (p \cdot r)^2 (\epsilon_{(IJ)} \epsilon_{(KL)}) \quad (16)$$

because ϵ itself is an antisymmetric tensor so it will become zero under the symmetrization, so that the second term is vanishing.

After some rearrangement of indices, we can obtain

$$\varepsilon_{IJ}^\mu \varepsilon_{KL,\mu} = \frac{1}{4}(\epsilon_{IL}\epsilon_{KJ} + \epsilon_{IK}\epsilon_{LJ}) \quad (17)$$

Unfortunately, when we want to reproduce the original normalization relation, we will obtain something like

$$\begin{aligned} \varepsilon_a^\mu \varepsilon_{b,\mu} &= (-\varepsilon_{(IJ)}^\mu (\sigma_a)^{IJ})(-\varepsilon_{(KL)}^\mu (\sigma_b)^{KL}) \\ &= \frac{1}{4}(\epsilon_{IL}\epsilon_{KJ} + \epsilon_{IK}\epsilon_{LJ})(\sigma_a)^{IJ}(\sigma_b)^{KL} \\ &= \frac{1}{4}\epsilon_{IL}\epsilon_{KJ}\epsilon^{JM}(\sigma_a)^I{}_M\epsilon^{LN}(\sigma_b)^K{}_N + \frac{1}{4}\epsilon_{IK}\epsilon_{LJ}\epsilon^{IM}(\sigma_a)^J{}_M\epsilon^{LN}(\sigma_b)^K{}_N \\ &= \frac{1}{4}\delta_I^N\delta_K^M(\sigma_a)^I{}_M(\sigma_b)^K{}_N + \frac{1}{4}\delta_K^M\delta_J^N(\sigma_a)^J{}_M(\sigma_b)^K{}_N \\ &= \frac{1}{4}2\text{tr}(\sigma_a\sigma_b) \\ &= \delta_{ab} \end{aligned} \quad (18)$$

2 Completeness Relation of Polarization Vectors

For this purpose, we need to compute the following formula

$$\varepsilon_a^\mu \varepsilon^{a,\nu} = ? \quad (19)$$

If we multiply the momentum p_μ , it should become vanishing so that the completeness relation should look like

$$\varepsilon_a^\mu \varepsilon^{a,\nu} = k(-\eta^{\mu\nu} + \frac{p^\mu r^\nu + p^\nu r^\mu}{(p \cdot r)^2}) \quad (20)$$

here k is just a overall coefficient.

Here, we choose to compute the completeness relation for polarization vectors with fundamental indices

$$\varepsilon_{IJ}^\mu \varepsilon^{IJ,\nu} = a(-\eta^{\mu\nu} + \frac{p^\mu r^\nu + p^\nu r^\mu}{(p \cdot r)^2}) \quad (21)$$

with a different coefficient.

For symplisity, we start with the convention from paper arxiv:2202.08257, in which the polarization vector looks like

$$\epsilon_{IJ}^\mu = \frac{\langle p_I | \Gamma^\mu | r_J \rangle}{\sqrt{2}} \quad (22)$$

with the normalization

$$\langle p^I | r^J \rangle = \epsilon^{IJ} \quad (23)$$

which means the following condition

$$2(p \cdot r) = 1 \quad (24)$$

The method to compute this completeness relation is still just multiplying them together like

$$\varepsilon_{(IJ)}^\mu \varepsilon^{(IJ),\nu} = \frac{\langle p_I | \Gamma^\mu | r_J \rangle}{\sqrt{2}} \times \frac{\langle p^I | \Gamma^\nu | r^J \rangle}{\sqrt{2}} \quad (25)$$

It can be decomposed into 4 pices IJ^{IJ} , IJ^{JI} , JJ^{IJ} , JJ^{JI} , and the first one can be computed like

$$\begin{aligned} \langle p_I | \Gamma^\mu | r_J \rangle \times \langle p^I | \Gamma^\nu | r^J \rangle &= -\langle p_I | \Gamma^\mu | r_J \rangle \langle r^J | \Gamma^\nu | p^I \rangle \\ &= -\text{Tr}[\langle p^I \rangle \langle p_I | \Gamma^\mu | r_J \rangle \langle r^J | \Gamma^\nu] \\ &= \text{Tr}[(p \cdot \Gamma) \Gamma^\mu (r \cdot \Gamma) \Gamma^\nu] \\ &= 4(p^\mu r^\nu + p^\nu r^\mu - (p \cdot r) \eta^{\mu\nu}) \end{aligned} \quad (26)$$

where we use a nontrivial fact that

$$\langle p^I | \Gamma^\nu | r^J \rangle = -\langle r^J | \Gamma^\nu | p^I \rangle \quad (\text{obtained from } C^T \Gamma^\mu C = -(\Gamma^\mu)^T) \quad (27)$$

The fourth piece is nearly the same as the first one

$$\langle p_J | \Gamma^\mu | r_I \rangle \times \langle p^J | \Gamma^\nu | r^I \rangle = 4(p^\mu r^\nu + p^\nu r^\mu - (p \cdot r) \eta^{\mu\nu}) \quad (28)$$

The remaining two need some extra efforts, here I compute it explicitly

$$\langle p_I | \Gamma^\mu | r_J \rangle \times \langle p^J | \Gamma^\nu | r^I \rangle \quad (29)$$

In terms of two massless spinors that satisfy (23), the $USp(2, 2)$ identity operator can be written as

$$|r_a\rangle_A \langle p^a|^B + |p_a\rangle_A \langle r^a|^B = \delta_A^B \quad (30)$$

It can be used to rewrite the equation (29) like

$$\begin{aligned} &\langle p_I |^A (\Gamma^\mu)_A^B |r_J\rangle_B \langle p^J |^C (\Gamma^\nu)_C^D |r^I\rangle_D \\ &= \langle p_I |^A (\Gamma^\mu)_A^B (\delta_B^C - |p_J\rangle_B \langle r^J|^C) (\Gamma^\nu)_C^D |r^I\rangle_D \\ &= \langle p_I |^A (\Gamma^\mu)_A^B (\Gamma^\nu)_B^D |r^I\rangle_D - \langle p_I | \Gamma^\mu | p_J \rangle \langle r^J | \Gamma^\nu | r^I \rangle \end{aligned} \quad (31)$$

And similarly for

$$\begin{aligned} \langle p_J | \Gamma^\mu | r_I \rangle \times \langle p^I | \Gamma^\nu | r^J \rangle &= \langle p^I | \Gamma^\nu | r^J \rangle \times \langle p_J | \Gamma^\mu | r_I \rangle \\ &= \langle p^I |^A (\Gamma^\nu)_A^B (-|p^J\rangle_B \langle r_J|^C - \delta_B^C) (\Gamma^\mu)_C^D |r_I\rangle_D \\ &= -\langle p^I |^A (\Gamma^\nu)_A^B (\Gamma^\mu)_B^D |r_I\rangle_D - \langle p^I | \Gamma^\nu | p^J \rangle \langle r_J | \Gamma^\mu | r_I \rangle \end{aligned} \quad (32)$$

Now we can combine these two terms

$$\begin{aligned} &\langle p_I |^A (\Gamma^\mu)_A^B (\Gamma^\nu)_B^D |r^I\rangle_D - \langle p_I | \Gamma^\mu | p_J \rangle \langle r^J | \Gamma^\nu | r^I \rangle - \langle p^I |^A (\Gamma^\nu)_A^B (\Gamma^\mu)_B^D |r_I\rangle_D - \langle p^I | \Gamma^\nu | p^J \rangle \langle r_J | \Gamma^\mu | r_I \rangle \\ &= \langle p_I | (\Gamma^\mu \Gamma^\nu + \Gamma^\nu \Gamma^\mu) | r^I \rangle - \langle p_I | \Gamma^\mu | p_J \rangle \langle r^J | \Gamma^\nu | r^I \rangle - \langle p^I | \Gamma^\nu | p^J \rangle \langle r_J | \Gamma^\mu | r_I \rangle \\ &= 2\eta^{\mu\nu} \langle p_I | r^I \rangle - (2p^\mu \epsilon_{IJ})(-2r^\nu \epsilon^{JI}) - (-2p^\nu \epsilon^{IJ})(2r^\mu \epsilon_{JI}) \\ &= 4\eta^{\mu\nu} + 8p^\mu r^\nu + 8p^\nu r^\mu \end{aligned} \quad (33)$$

Above all, we can conclude that

$$\begin{aligned} \varepsilon_{(IJ)}^\mu \varepsilon^{(IJ),\nu} &= \frac{1}{2} \times \frac{1}{8} [8(p^\mu r^\nu + p^\nu r^\mu - (p \cdot r) \eta^{\mu\nu}) + 4\eta^{\mu\nu} + 8p^\mu r^\nu + 8p^\nu r^\mu] \\ &\stackrel{?}{=} 2p^{(\mu} r^{\nu)} \end{aligned} \quad (34)$$